

{EPITECH}

<SINCE 1999/>

< Robotics TEK5 />

Kickoff - Projet : LeRobot

{ <TECH TOGETHER TOMORROW> }



	Groupe A - Octobre	Groupe B - Fin Novembre
Kickoff + Bootstrap	8 et 9 Octobre	26 et 27 Novembre
<i>Cours Visio #1</i>	22 Octobre	10 Décembre
Sout. Interm.	~31 Octobre	~19 Décembre
<i>Cours Visio #2</i>	5 Novembre	7 Janvier
Sout. Finale	21 ou 28 Novembre	23 ou 30 Janvier

{EPITECH}

<SINCE 1999/>

< Robotics TEK5 />

The *Evolution* of **Robotics**: From Early **Automation** to Modern **AI**

{ <TECH TOGETHER TOMORROW> }



< From Automation ... />



Karakuri Ningyo (18th-19th century)



Le métier Jacquard (1804)



The androids from Jaquet-Droz (1774)

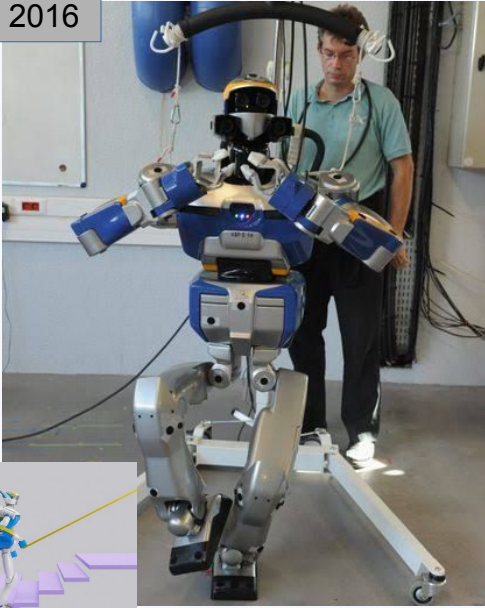
[Video](#)

< ... to Modern Robotics />

Golden age of « Classical Control »

Before ~2015

2016



2005



2010



2008 -
2014



Video LAAS-CNRS

Golden age of « Classical Control »
Before ~2015

2016



2005



2010



2008 -
2014

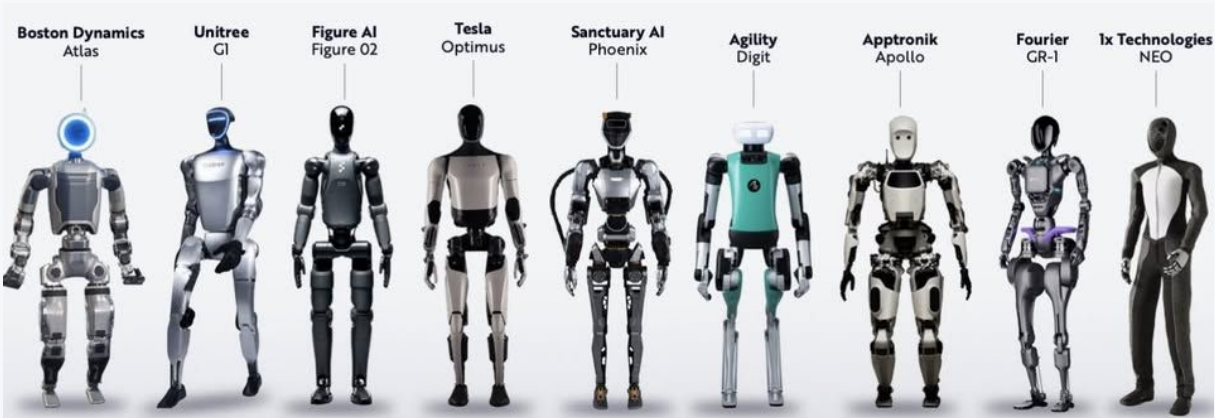


Video LAAS-CNRS

Modern & Adaptative Robotics
Today (2021-2026)



Video Nouvel An Chinois 2026



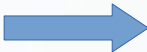
< The Rise of AI />

1980s – 2000s

ML Theory



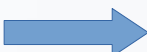
Y. LeCun
CNN



Deep learning Y LeCun, Y Bengio, G Hinton nature 521 (7553), 436-444	123876	2015
Gradient-based learning applied to document recognition Y LeCun, L Bottou, Y Bengio, P Haffner Proceedings of the IEEE 86 (11), 2278-2324	90442	1998
Backpropagation applied to handwritten zip code recognition Y LeCun, B Boser, JS Denker, D Henderson, RE Howard, W Hubbard, ... Neural computation 1 (4), 541-551	20766	1989
Convolutional networks for images, speech, and time series Y LeCun, Y Bengio The handbook of brain theory and neural networks	10147	1998
The MNIST database of handwritten digits Y LeCun, C Cortes	9892	1998



R. Sutton
RL



Reinforcement learning: An Introduction, 2nd edition RS Sutton, AG Barto MIT press	102122	2018
Policy gradient methods for reinforcement learning with function approximation RS Sutton, D McAllester, S Singh, Y Mansour Advances in neural information processing systems 12	11629	1999
Learning to predict by the methods of temporal differences RS Sutton Machine learning 3 (1), 9-44	9488	1988
Reinforcement learning: An Introduction, 1st edition RS Sutton, AG Barto MIT press	6562 *	1998
Between MDPs and semi-MDPs: A framework for temporal abstraction in reinforcement learning	5990	1999

< The Rise of AI />



1980s – 2000s

2007 – 2012

ML Theory

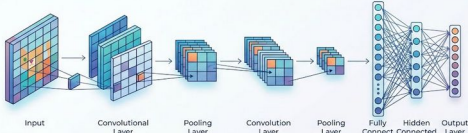


Y. LeCun
CNN



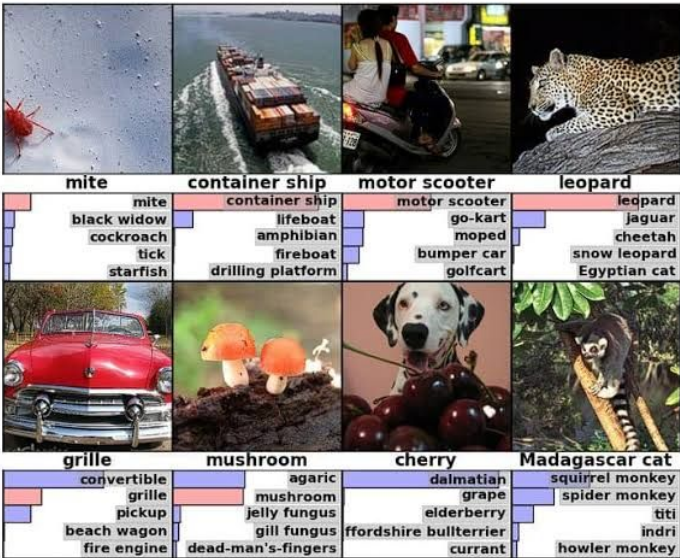
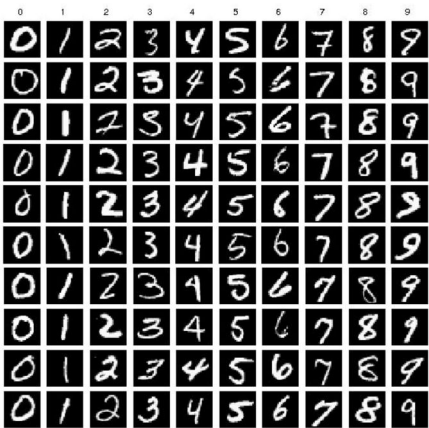
R. Sutton
RL

The Hardware Breakthrough:
CUDA & GPU-Powered
Deep Learning

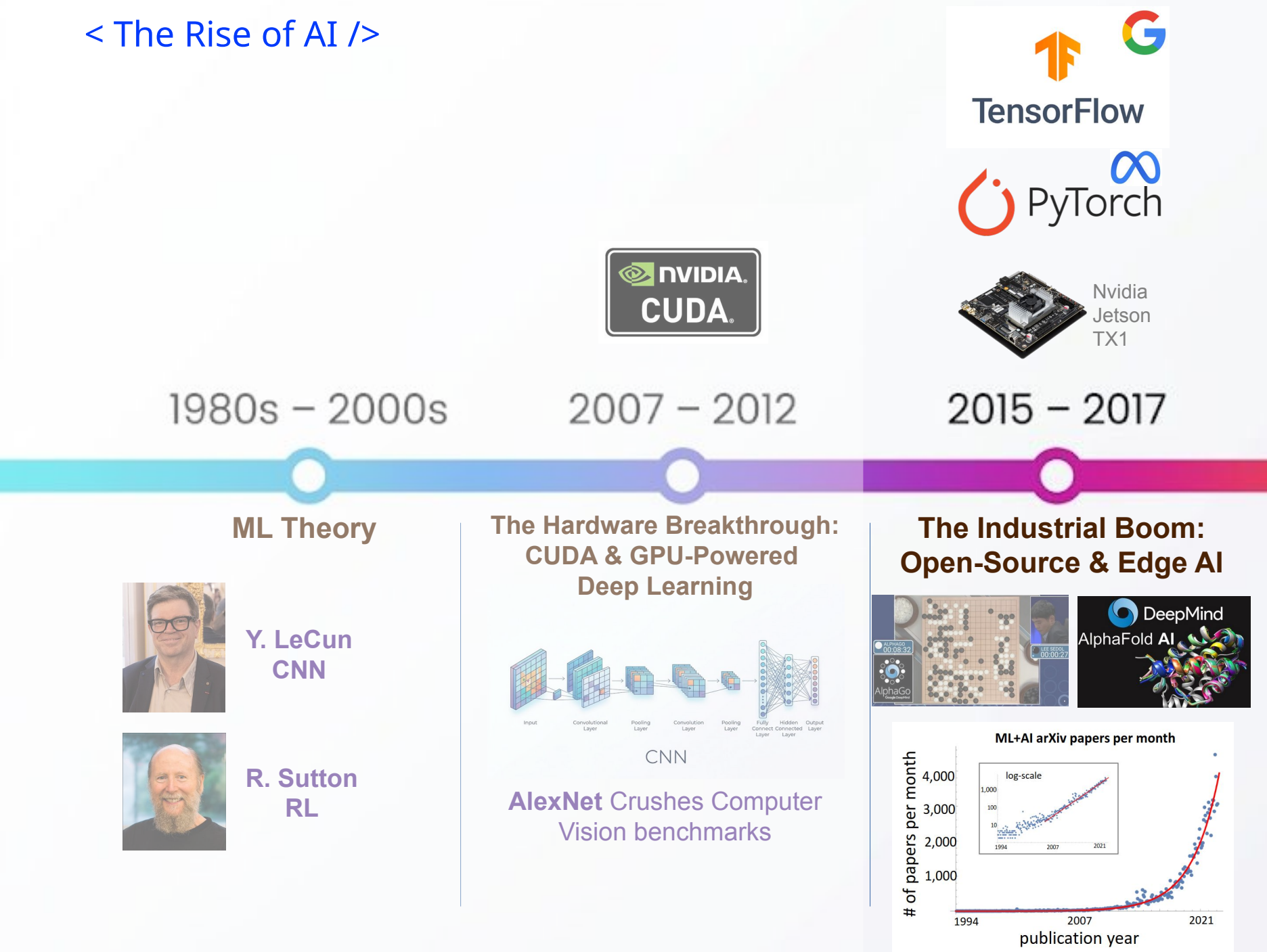


CNN

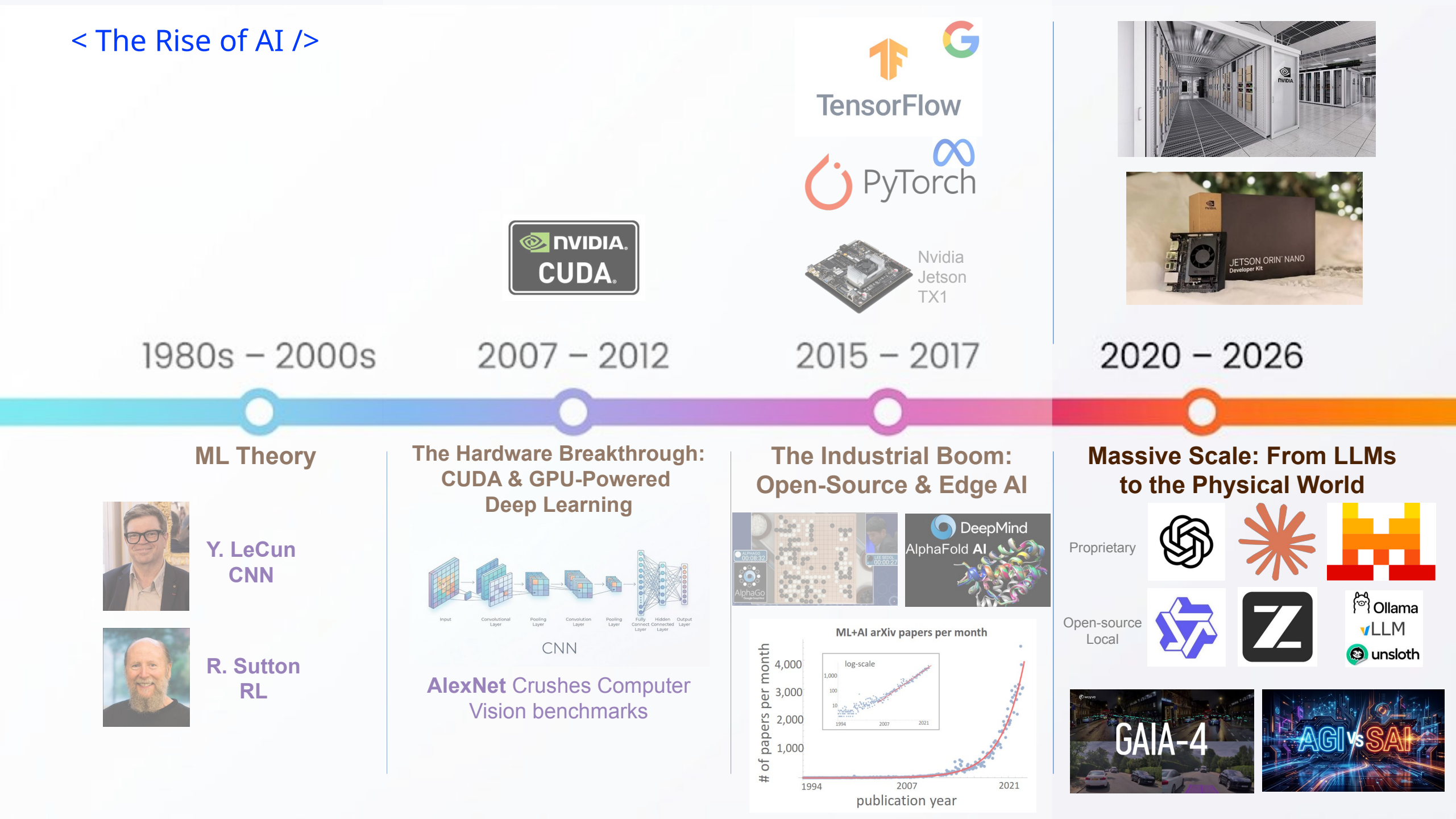
AlexNet Crushes Computer
Vision benchmarks



< The Rise of AI />



< The Rise of AI />

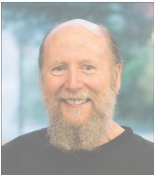


1980s – 2000s

ML Theory



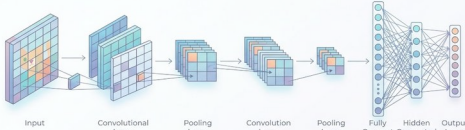
Y. LeCun
CNN



R. Sutton
RL

2007 – 2012

The Hardware Breakthrough:
CUDA & GPU-Powered
Deep Learning



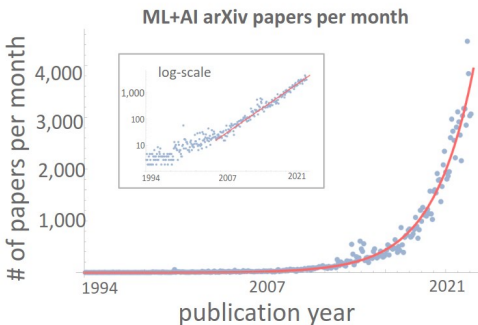
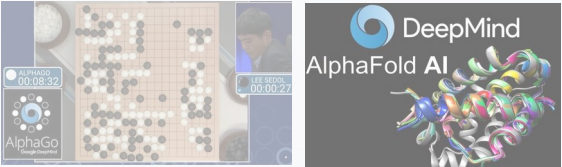
CNN

AlexNet Crushes Computer
Vision benchmarks



2015 – 2017

The Industrial Boom:
Open-Source & Edge AI



2020 – 2026

Massive Scale: From LLMs
to the Physical World



< The Rise of AI in Robotics />

From Classical Control to AI & Hybrid approaches

Model Predictive Control (MPC)



🎯 Move to target with shortest path

🔗 No collision + Motor limits

⚡ Optimise on horizon H
Apply action u_0
Recompute

Reinforcement Learning (RL)



« Yesterday we tuned cost functions in MPC,
today we tune rewards in RL: the job is pretty
much the same. » N. Mansard

< The Rise of AI in Robotics />

From Classical Control
to AI & Hybrid approaches

Model Predictive Control (MPC)

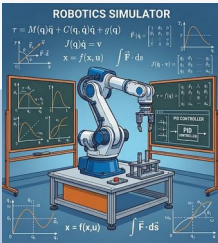


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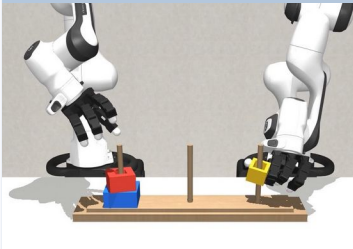
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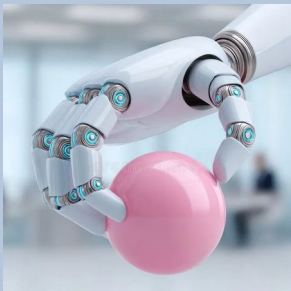
Sim2Real



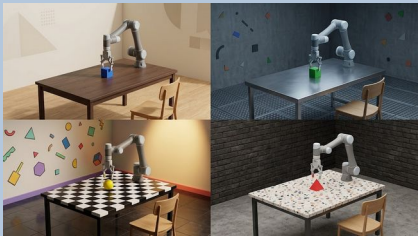
Mujoco (2012)



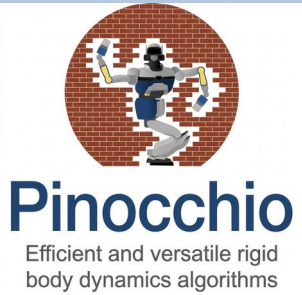
Isaac Sim (2018)



The Reality Gap: Contact Dynamics, Friction ...



Domain Randomization



< The Rise of AI in Robotics />

From Classical Control to AI & Hybrid approaches

Model Predictive Control (MPC)

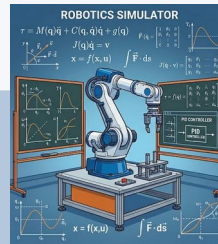


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- Apply action u_0
- Recompute

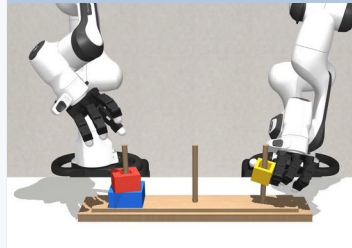
Reinforcement Learning (RL)



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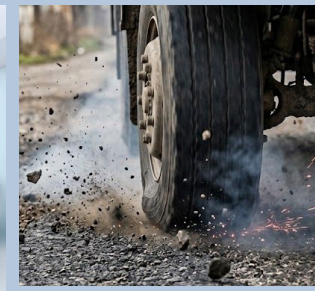
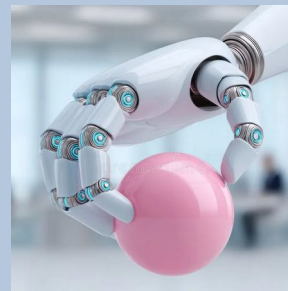
Sim2Real



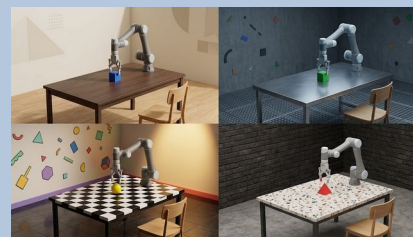
Mujoco (2012)



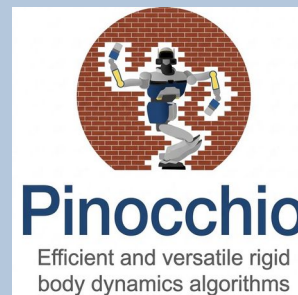
Isaac Sim (2018)



The Reality Gap: Contact Dynamics, Friction ...



Domain Randomization



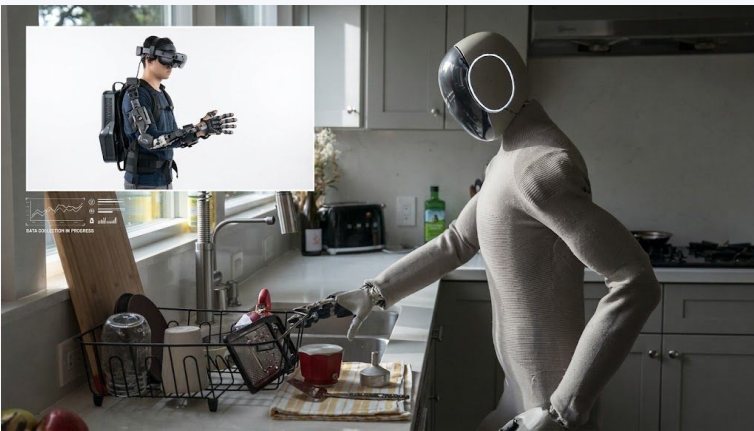
Future : World models & Physical AI ?

What if the simulator becomes a neural network ?



< World Models : Scaling Real-World Data />

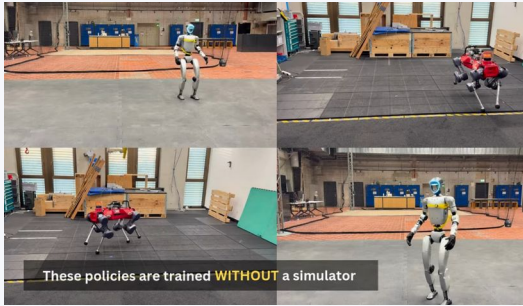
Data Collection bottleneck



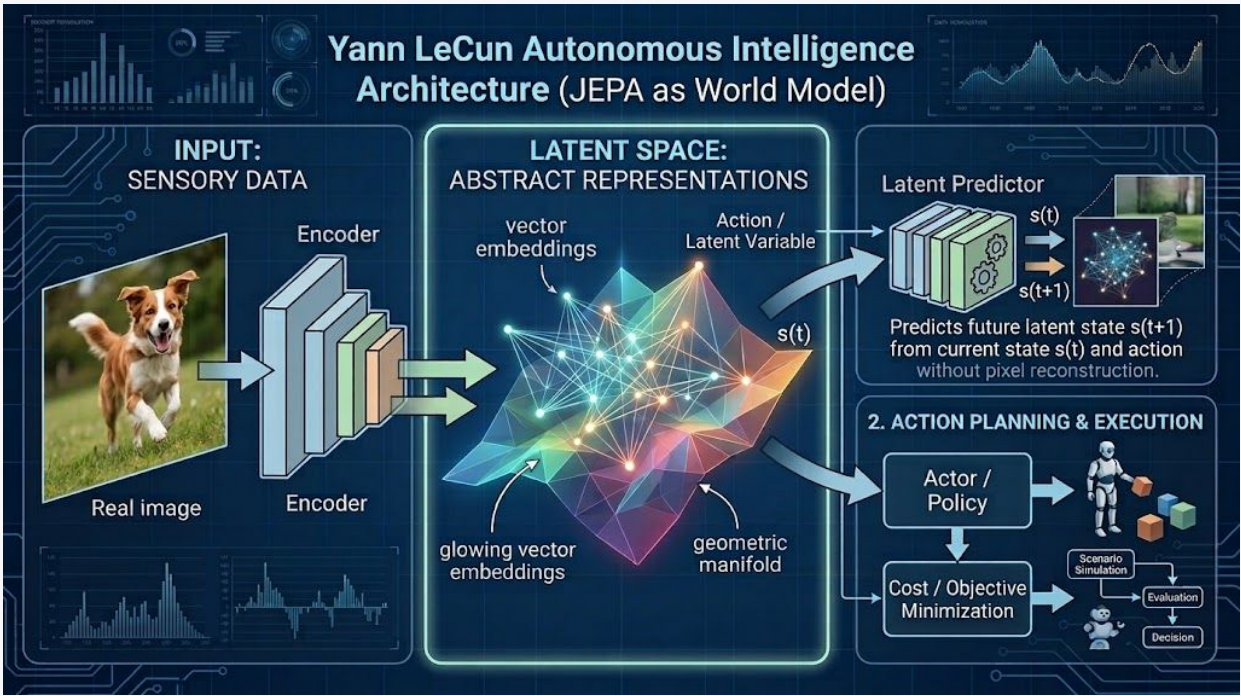
Google Deepmind
Generative Vision



Wayve
Generative Vision (Predict the world into pixels)
<https://wayve.ai/thinking/gaia-4/>



ETH Zurich
[Video](#)



{EPITECH}

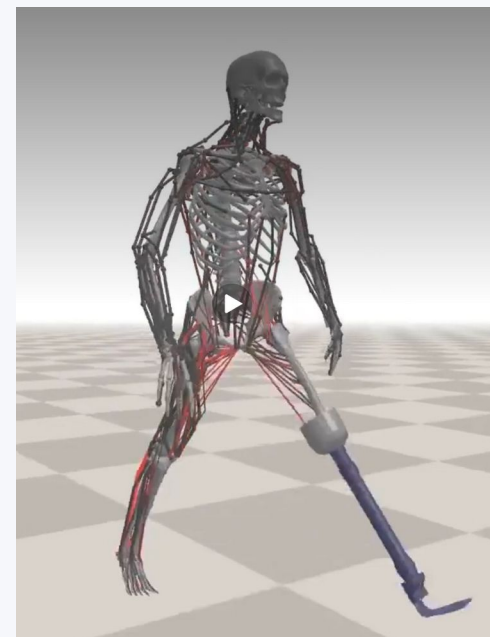
<SINCE 1999/>

< Robotics TEK5 />

Where I Stand on This Timeline

{ <TECH TOGETHER TOMORROW> }

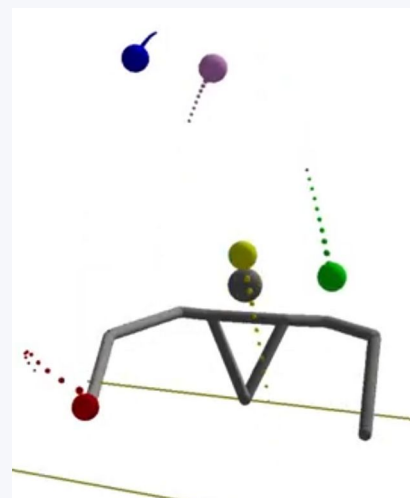




Muscles + RL
2019



"Soft Shadow Art"
2017



Juggling + RL
2018



"How to train your dragon" (RL)
2017

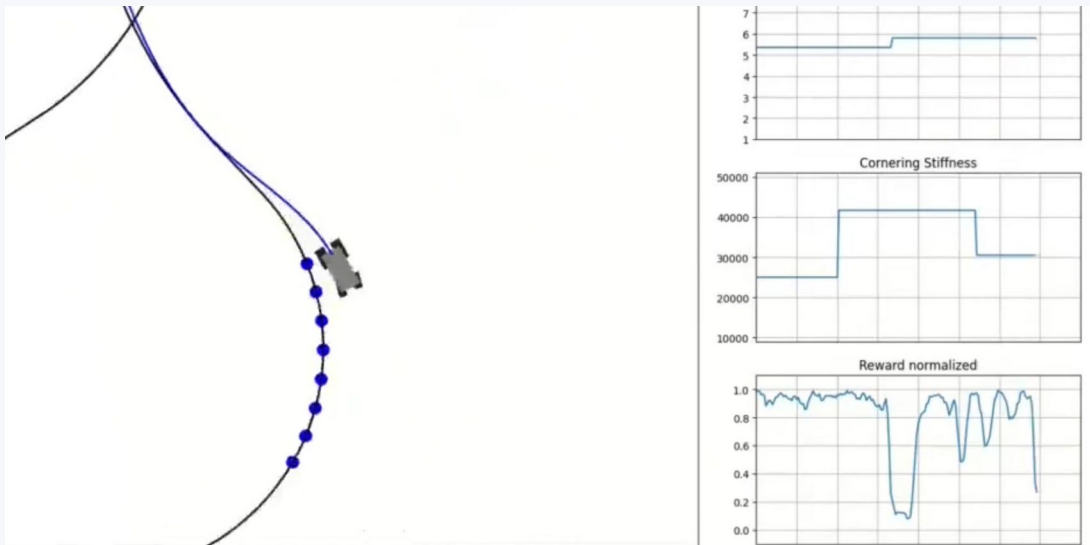


PhD Thesis : “Reinforcement learning of a navigation method for contact planning on humanoid robots” 2022

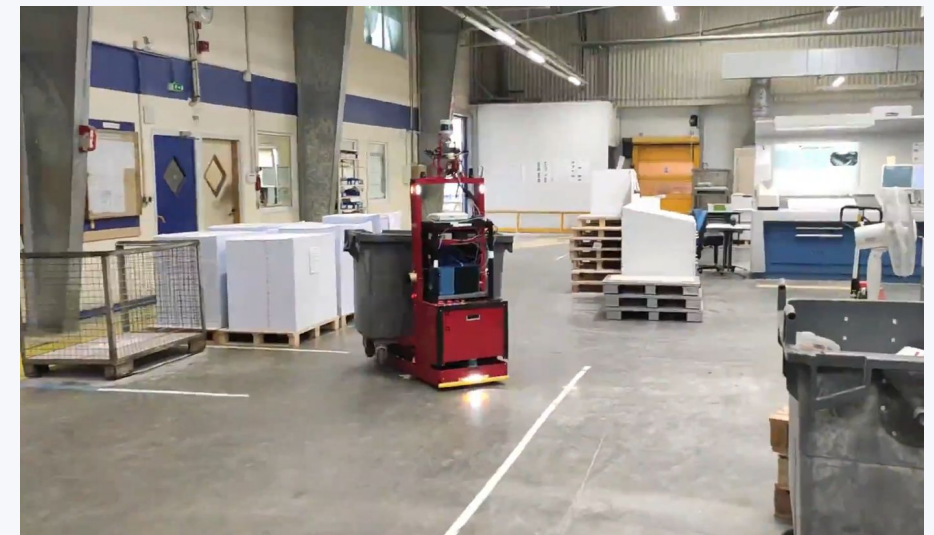


Video





“A Study of Reinforcement Learning Techniques for Path Tracking in Autonomous Vehicles” 2024



“Autonomous forklift navigation inside a cluttered logistics factory” 2024

2012

INSA
TOULOUSE

2016

서울대학교
SEOUL NATIONAL UNIVERSITY

2019

LAAS
CNRS

2022

cea list

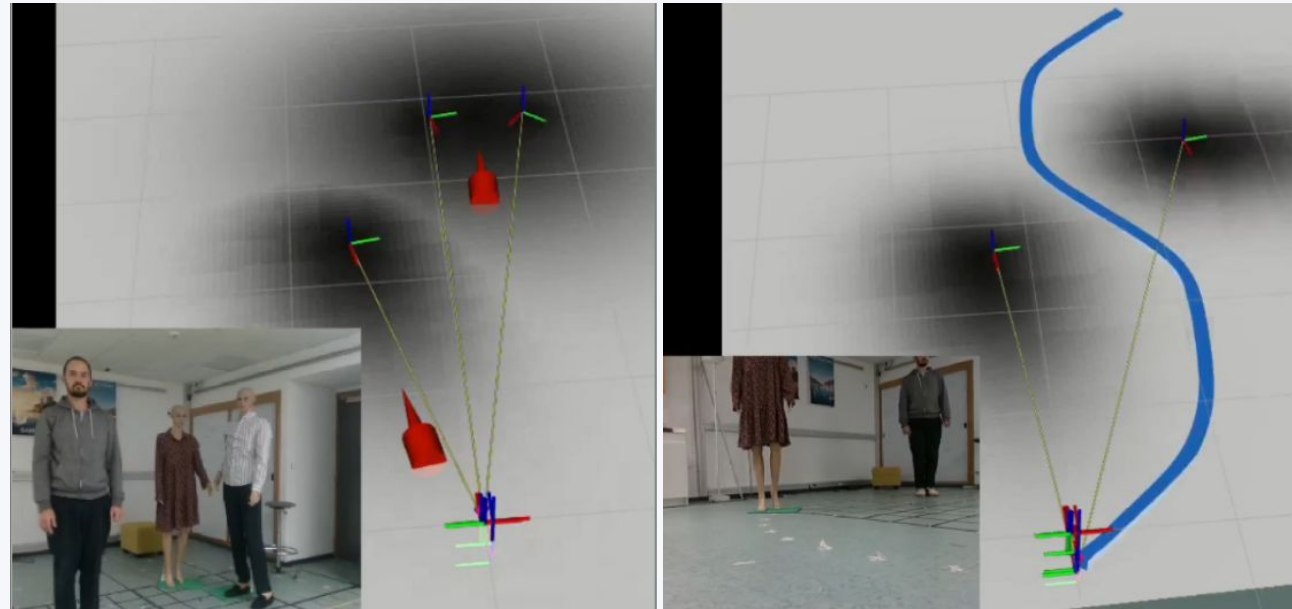
2025

2026

Inria



Mirokai Robot
(Enchanted Tools - Paris)
Video



Social navigation between Humans and Robots

ML Theory
+
The Hardware Breakthrough:
CUDA & GPU-Powered Deep Learning

The Industrial Boom:
Open-Source & Edge AI

Massive Scale: From LLMs
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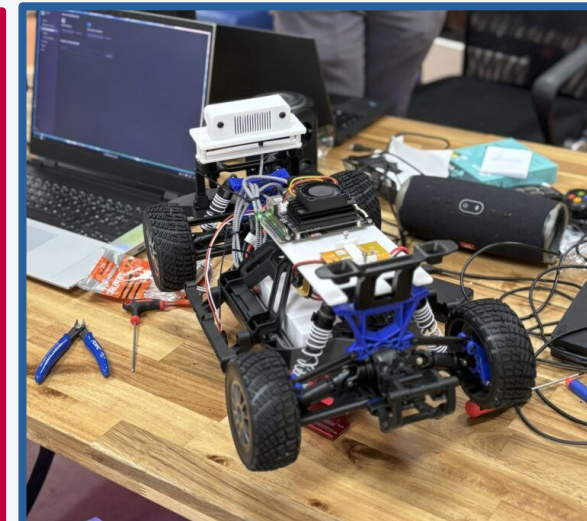


Our Robotics Software Stack

- **Embedded & Middleware**
Python/C++ | embedded Linux | ROS 2 | DDS/CAN | Real-time
- **Perception & Computer Vision**
Camera | LiDAR | Sensor fusion | Object detection | SLAM
- **Autonomous Navigation**
Pathfinding | Obstacle avoidance | Mobile kinematics
- **Control & Manipulation**
inverse kinematics | trajectory planning | feedback control
- **AI**
Machine learning | Reinforcement learning | VLM/LLM | Edge AI

EPITECH

Depuis juin 2026



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<SINCE 1999/>

< Robotics TEK5 />



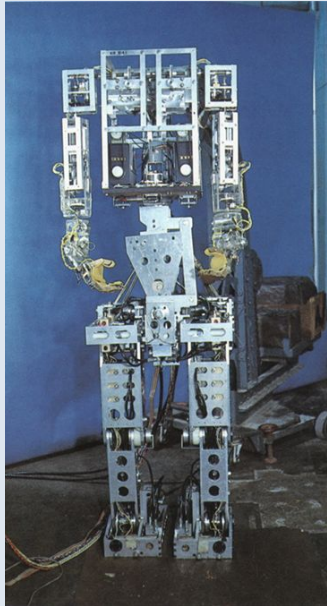
The Engineering Reality: The Modern Robotic Stack

{ <TECH TOGETHER TOMORROW> }



< The Middleware Backbone: From Chaos to Industry />

Before 2007



WABOT1 (1973)



Honda P2 (1996)

Chaos & Isolation

Zero software unification,
every lab builds from scratch

ROS (2007) by Willow Garage



Robots PR2 distribués aux plus grands labo de
recherche (MIT, Stanford, TUM, LAAS-CNRS)

The Open-Source Standard

Modular nodes, message passing
(pub/sub), and unifying robotics
research worldwide.

ROS2 (2017+) maintained by Open Robotics

Founders of willow + Alphabet
Supervised by : AWS, Nvidia, Sony,
Intel, Toyota research...

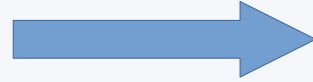


From R&D to the Industrial Standard

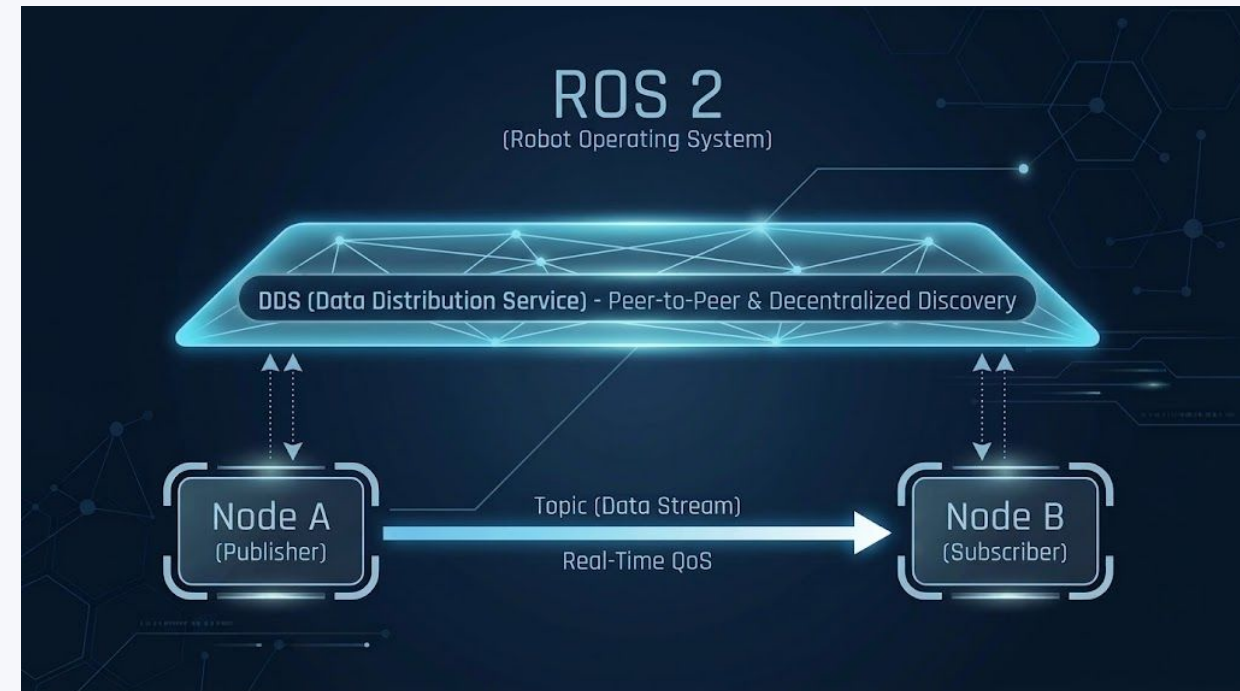
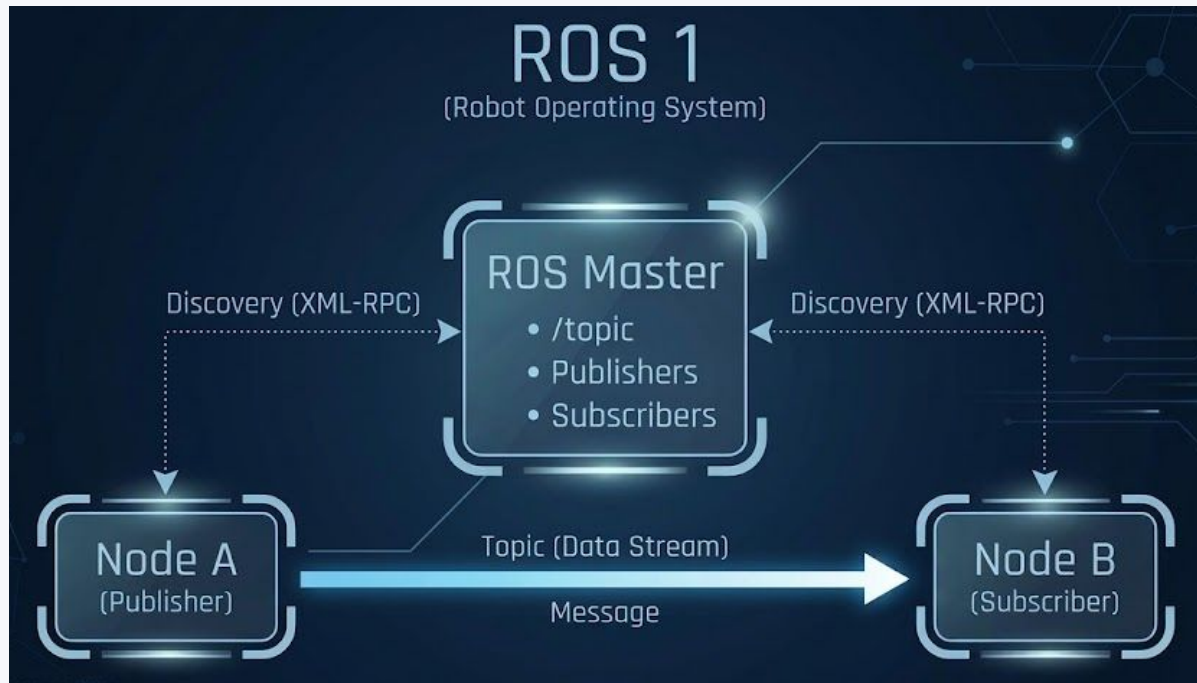
Real-time distributed architecture,
production-ready, and adopted from
space to factory floors.

< Architectural Shift: Why ROS 2 ? />

ROS



ROS2

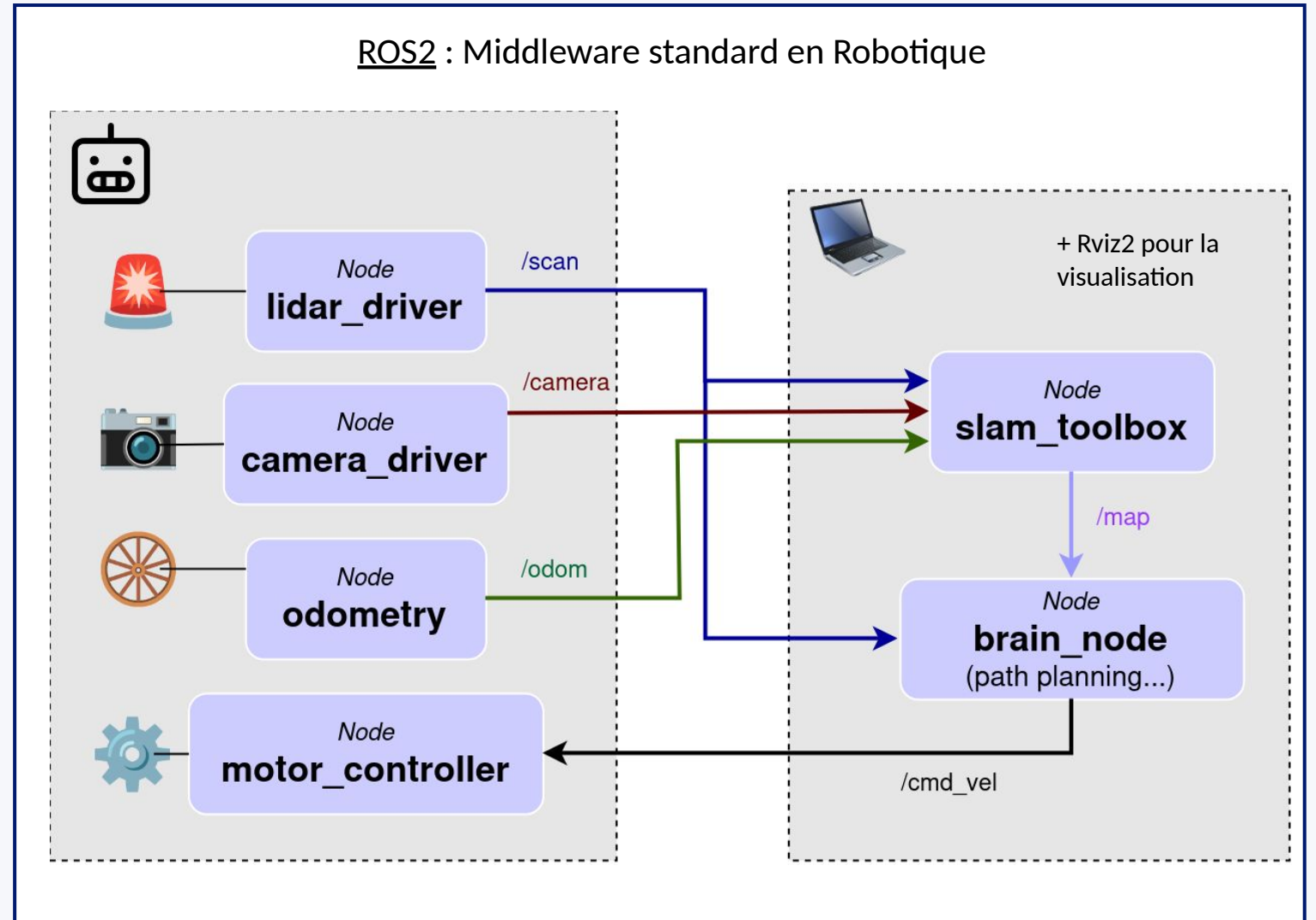


< ROS2 />

Node : Séparation des programmes

Topics : Canaux pour publier ou recevoir des datas (Publisher or Subscriber)

Réseau DDS (Data Distribution Service) :
Protocole réseau décentralisé
Transporte la data entre les nodes



ROS2



2022 – 2027
Ubuntu 22.04



2024 – 2029
Ubuntu 24.04



2026 – 2031
Ubuntu 26.04

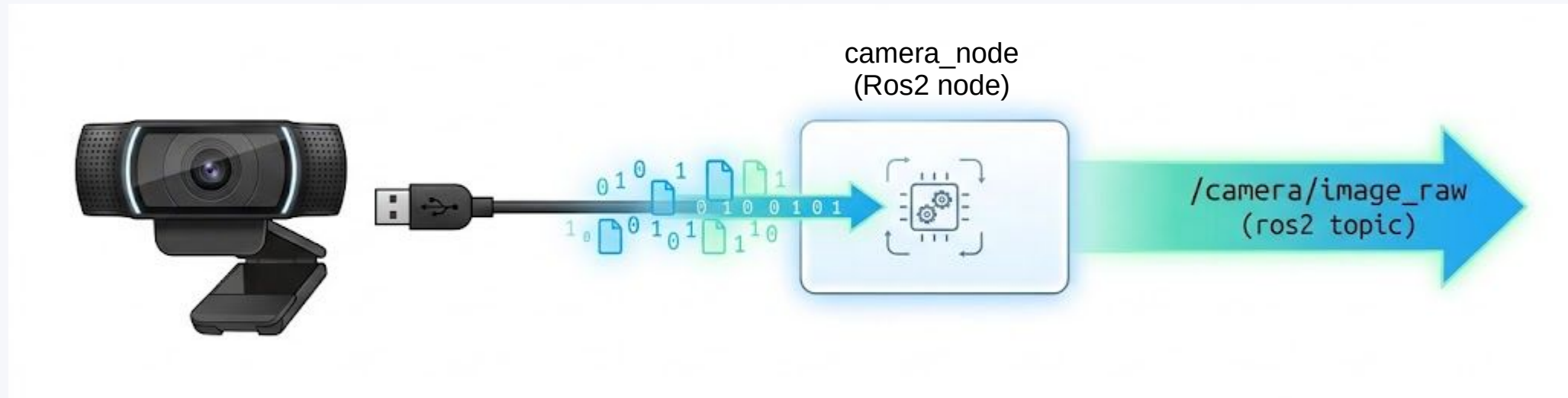
News:

- C++17
- Reinforced CycloneDDS & Zenoh
- Stricer QoS

News:

- C++20
- Deep Zenoh integration
- AI pipeline

< Key ROS2 commands />



Is my node running ?

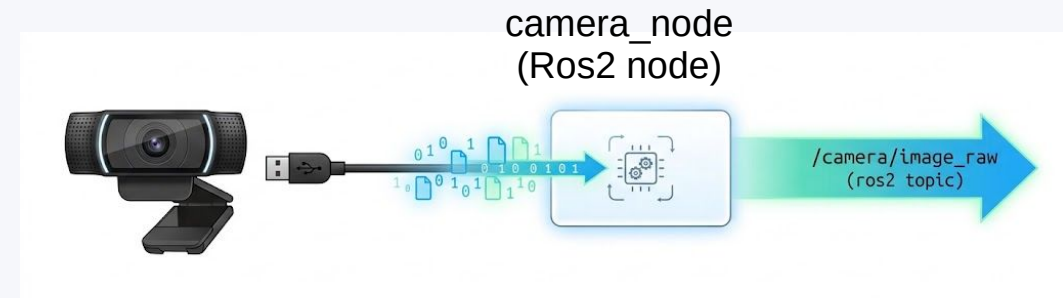
`ros2 topic hz /camera/image_raw`

*Is my node publishing
to the camera topic ?*

```
alex@robot-station:~/workspace$ ros2 node list
/camera_node ←
/laser_scanner
/map_server
/robot_state_publisher
/rviz2_node
/teleop_twist_keyboard
/transform_publisher
/twist_mux
```

```
alex@robot-station:~/workspace$ ros2 topic list
/camera/camera_info
/camera/image_raw ←
/camera/depth/image_raw
/joint_states
/map
/odom
/scan
/tf
/tf_static
/teleop/cmd_vel
/parameter_events
/rosout
```

< Key ROS2 commands : **echo** />



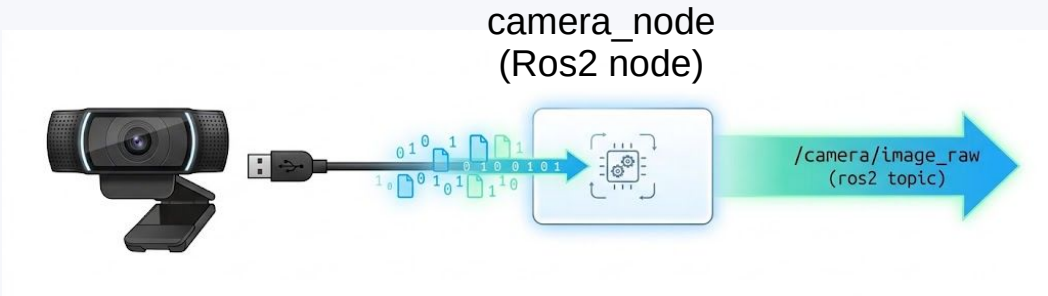
Am I receiving messages from this topic ?

```
alex@robot-station:~/workspace$ ros2 topic echo /camera/im
age_raw
header:
  stamp: sec: 1698330935
        nanosec: 498100000
  frame_id: camera_link
height: 480
width: 640
encoding: rgb8
is_bigendian: 0
step: 1920
data: [255, 255, 255, 255, 255, 255, ..., 255]
```

< Key ROS2 commands : **hz** />

At what frequency ?

```
alex@robot-station:~/workspace$ ros2 topic hz /camera/i
mage_raw
average rate: 4.981
min: 0.198s max: 0.203s std dev: 0.002s window: 50
```



Typical Rates in Robotics

 High-Level / Heavy Data 1 to 10 Hz	 Global Path Planning ('/plan')	1 – 5 Hz	Re-calculer une route complète n'a pas besoin d'être instantané.
	 LiDAR scans ('/scan')	10 Hz	Vitesse de rotation physique du capteur.
	 AI Detection / Segmentation	5 – 15 Hz	Limité par la puissance de calcul (GPU/NPU).
 Perception & Video 15 to 60 Hz	 Standard Camera stream	30 Hz	Standard pour suivi visuel fluide / V-SLAM.
	<i>Note : à 5 Hz (comme sur le terminal), c'est soit de la détection lourde, soit un goulot d'étranglement USB/driver.</i>		
 Low-Level Control 50 to 1000 Hz	 Mobile Base Velocity ('/cmd_vel')	20 – 50 Hz	Réactivité sans saturer la liaison série.
	 Joint States & Arm Feedback	50 – 100 Hz	Suivi de trajectoire fluide.
	 Motor Current / Torque Loops	1000 Hz (1 kHz)	Boucle temps réel dur (microcontrôleur/CAN).

< Key ROS2 commands : **info** & QoS />

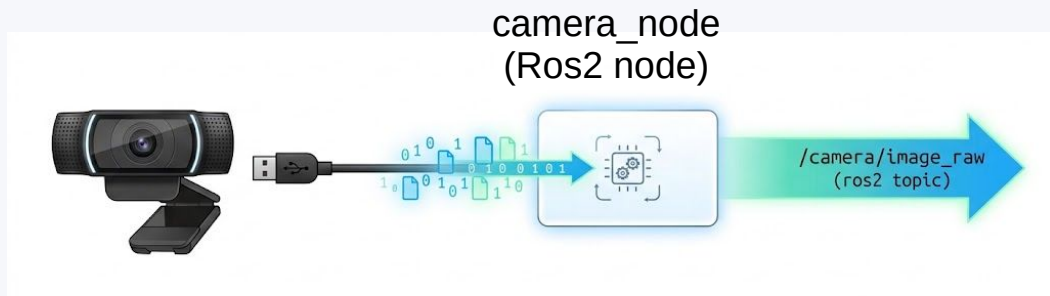
*Who is publishing, subscribing ?
What is the QoS profile ?*

```
alex@robot-station:~/workspace$ ros2 topic info -v /camera/image_raw
```

```
Topic: /camera/image_raw
Type: sensor_msgs/msg/Image
Publisher count: 1
Subscriber count: 1

Node name: camera_node
Node namespace: /
Topic type: sensor_msgs/msg/Image
Endpoint type: PUBLISHER
GID: 01.0f.a4...
QoS profile:
  Reliability: BEST_EFFORT
  Durability: VOLATILE
  Lifespan: Infinite
  Deadline: Infinite
  Liveliness: AUTOMATIC
```

```
Node name: image_viewer
Node namespace: /
Topic type: sensor_msgs/msg/Image
Endpoint type: SUBSCRIPTION
GID: 01.0f.b2...
QoS profile:
  Reliability: BEST_EFFORT
  Durability: VOLATILE
  Lifespan: Infinite
  Deadline: Infinite
  Liveliness: AUTOMATIC
```



Reliability (!!! Critical)

- **Best Effort** : ~UDP → Heavy continuous streams (camera, LiDAR ...)
- **Reliable** : ~TCP → Critical commands (emergency stop ...)

A reliable subscriber wants some guarantees on the QoS.
A best effort can not ensure these guarantees.
=> Zero message received (and no error, silent).

Durability

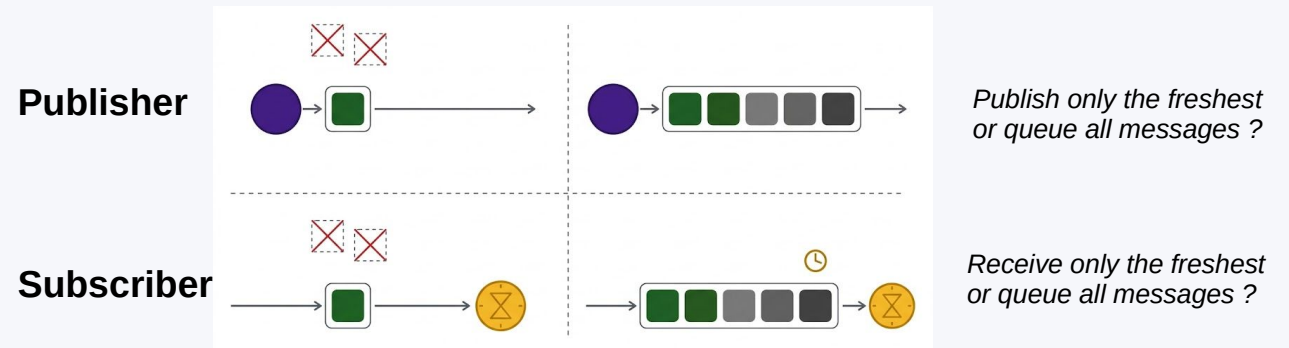
- **Volatile** : No history kept
- **Transient Local** : Publisher keep the last message for late-joining subscribers

Other policies

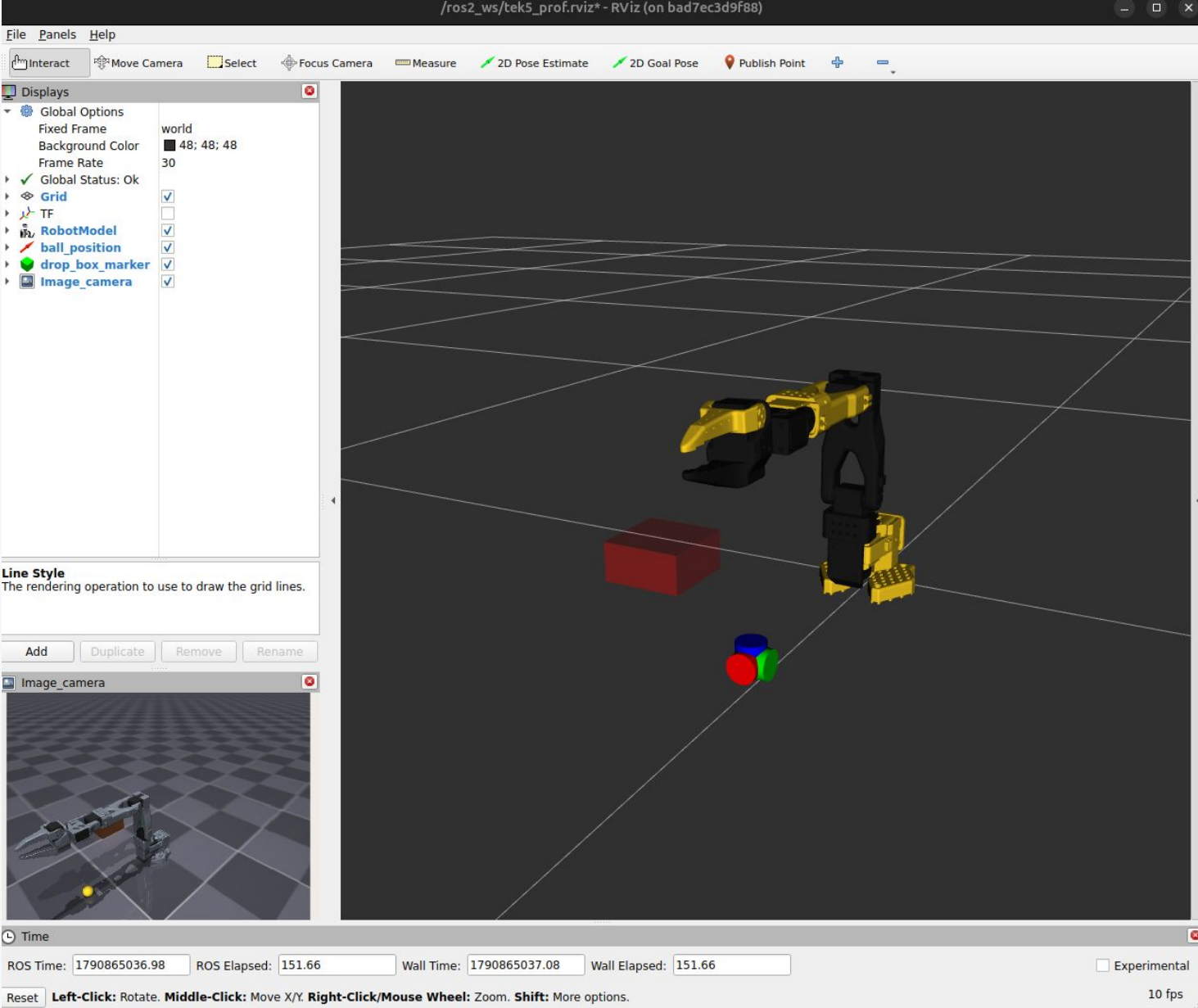
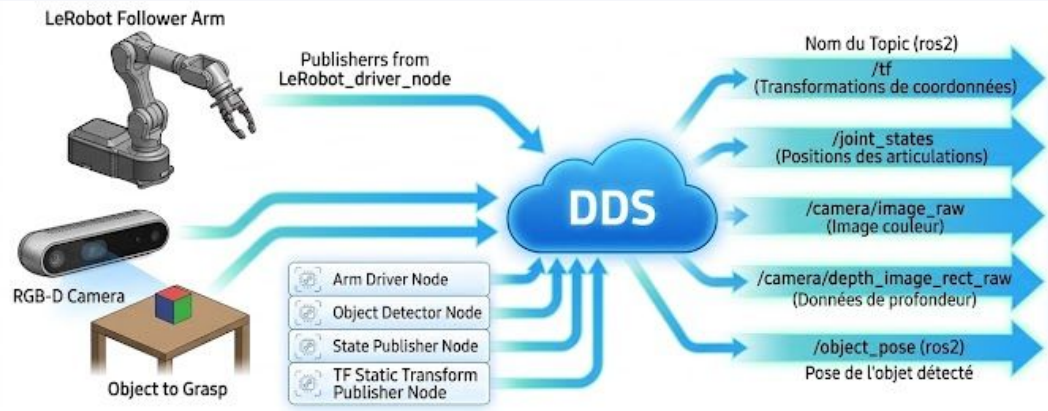
- **Lifespan** : Message expiration timeout
- **Deadline** : Maximum expected delay between publications
- **Liveliness** : Node health

Queue Depth

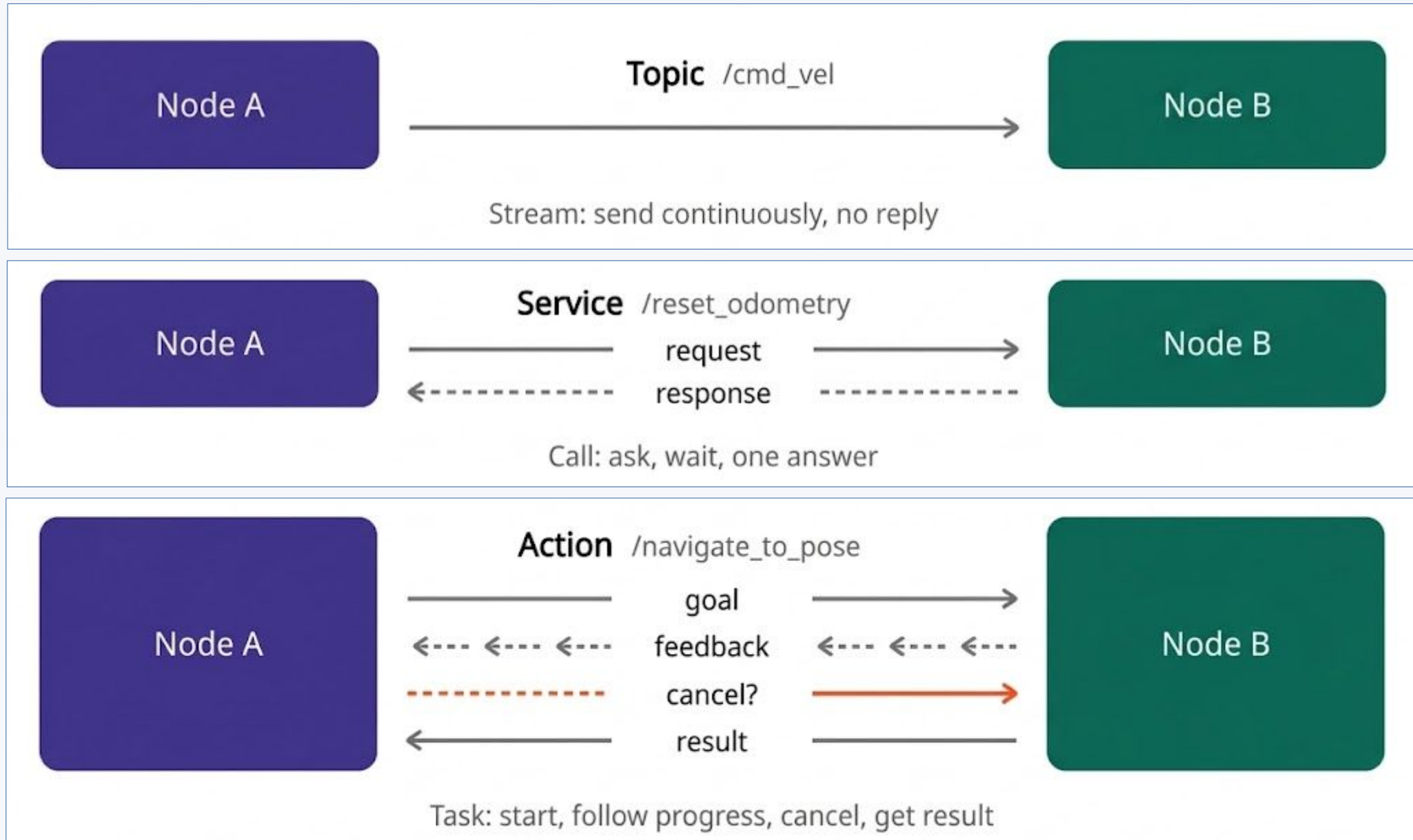
```
self.create_publisher(Image, '/camera/image_raw', 1)
```



< Visualisation : Rviz2 />



< ROS 2 communication : Topic, Service or Action ? />



< Build system : Colcon />

```
source /opt/ros/jazzy/setup.bash # ROS 2 itself, in every new terminal
colcon build                     # build all packages in src/
source install/setup.bash       # your packages, after each build
ros2 run <package> <node>       # run a node
```

```
ros2_ws/
├── src/                ← you write here
│   ├── my_robot_pkg/
│   │   ├── package.xml
│   │   ├── CMakeLists.txt
│   │   └── src/
│   └── my_nav_pkg/
├── build/              ← generated by colcon
├── install/            ← generated, source setup.bash
└── log/                ← generated, build logs
```

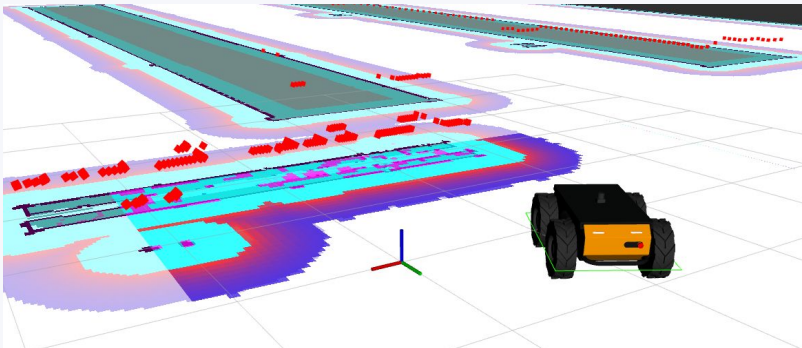
In the project

→ Everything runs in pre-built Docker containers.

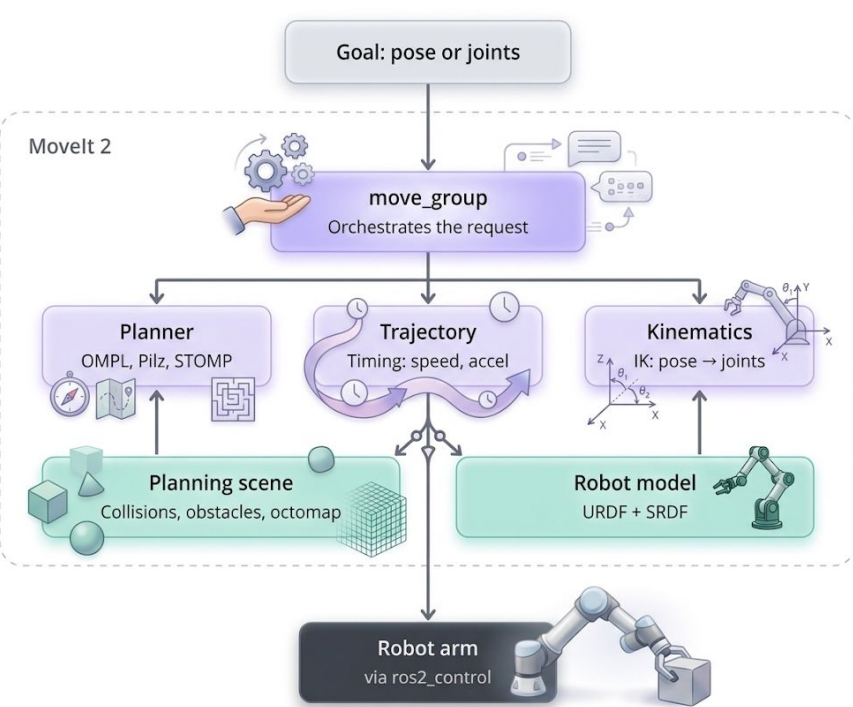
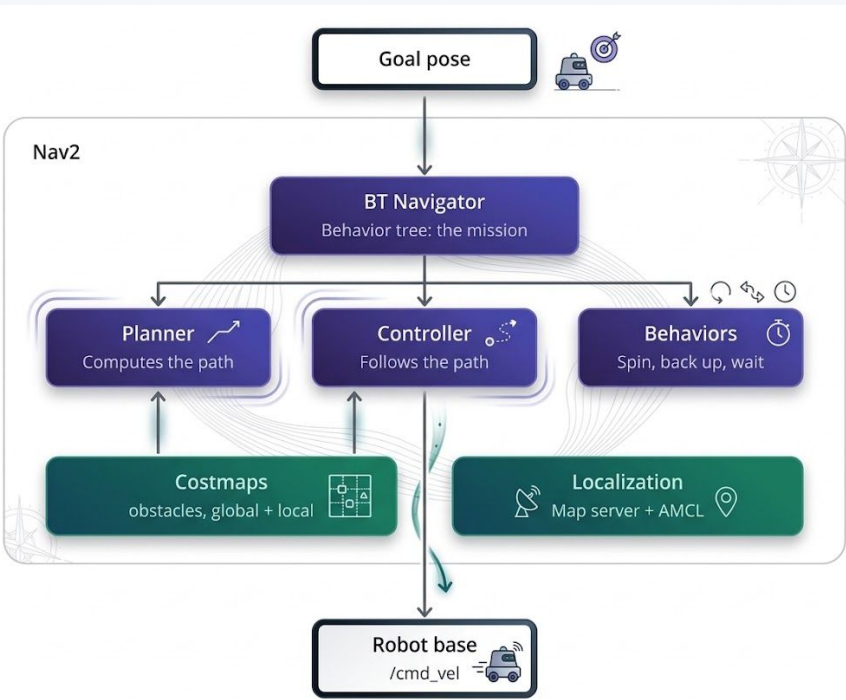
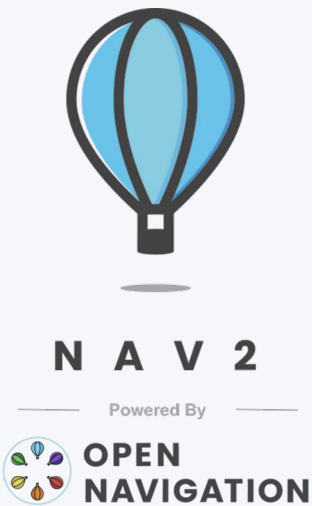
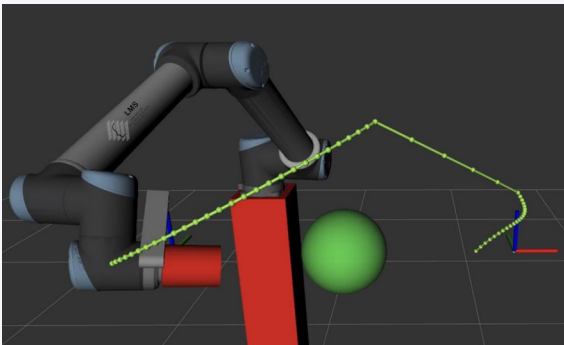
Your job

→ Adapt the code and launch the right containers.

< ROS 2 packages : Nav2 & MoveIt2 />



Video darpa



MoveIt 2 Software Architecture and Manipulation Flow